Trustworthy Collective Reasoning and Evaluation for AI Agents

Xinya Du, The University of Texas at Dallas (xinya.du@utdallas.edu)

ONR FY27 YIP white paper for Dr. Predrag Neskovic, Code 311, Mathematical Data Science

1 The problem

Five AI agents read shipping reports and are asked whether a cargo vessel has left port. Three say yes and two say no, but the vessel is still in port. Two of the yes answers rest on the same outdated schedule, and the third comes from a compromised agent that argues confidently for yes. A majority vote picks the wrong answer, and open discussion may do no better, because the compromised agent is persuasive and the two agents with the same source back each other up.

The central problem of this project is how a group of AI agents of unknown and correlated reliability, some possibly dishonest, can reach trustworthy decisions.

The example shows three ways collective judgment fails: agents that share a mistake count as independent votes, a dishonest agent can argue (or rate others) strategically, and discussion rewards confidence and persistence, with or without evidence. Verifying every claim would prevent these failures, but trustworthy verification cannot rest on the agents' own judgment. It takes a person (such as an analyst or an expert) or a direct measurement (such as a replication or a satellite image). Both are slow and costly, and as AI agents produce more claims, the share people can verify only shrinks. An AI verifier alone does not help: it is one more agent of unknown reliability.

2 Why it matters

AI models now grade one another's work, and multi-agent systems decide by vote or discussion (Zheng et al., 2023; Verga et al., 2024). Their errors are correlated, more so as models improve (Kim et al., 2025; Goel et al., 2025), so their agreement is weaker evidence than it looks. In one study, nine AI judges carried the information of about two independent ones, and items all nine agreed on were still wrong 9.1% of the time (Kohli, 2026). Debate makes agreement less informative: one round raised a panel's unanimity from 39.5% to 95.2% with little change in accuracy (Shu, 2026). Agents also collude. Pairs of agents that verified each other's work colluded in 94% of runs across ten models when breaking a rule paid more than following it (Shi et al., 2026).

Navy decisions face all of these problems at once: AI tools differ in reliability that is rarely known in advance, the true answer often arrives late, an adversary may compromise an agent or plant false reports, and verification is costly or impossible when connectivity is limited. A commander must still decide whether to act on what the agents agree on.

3 How it is solved today, and the limitations

Statistical methods estimate each rater's accuracy without an answer key (Dawid and Skene, 1979; Whitehill et al., 2009) and detect sources that copy one another (Dong et al., 2009), but they assume that raters err independently or in simple patterns (Jaffe et al., 2016) and never change their minds. Two recent preprints estimate the accuracy and shared mistakes of AI judges from small trusted sets (Hossain et al., 2026; Zhu et al., 2026), but treat the judges as honest and as never conferring.

Computational social choice designs impartial peer review, in which a participant's ratings cannot

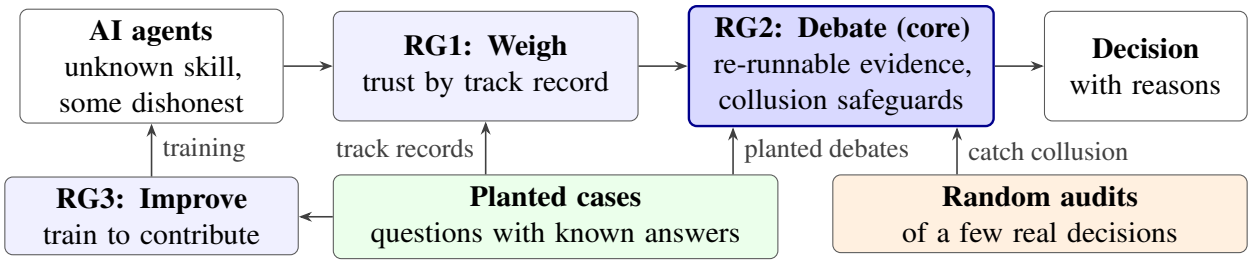


Figure 1: RG2, the core, settles disagreements by agentic debate with re-runnable evidence. Planted cases supply track records, planted debates, and training data; planted debates and random audits catch collusion.

change its own chance of selection (Kurokawa et al., 2015; Alon et al., 2011), and fair panel selection (Flanigan et al., 2021). Lev et al. (2023) added reliability weights to impartial selection, but their weights reward agreeing with the majority, and the authors note that accurate reviewers can no longer be identified once they are half or fewer. These mechanisms take opinions as given; in our setting some claims can be verified, and opinions change as agents talk.

Letting models debate can help (Du et al., 2024), but majority voting explains most of the gain (Choi et al., 2025), and a single persuasive adversary can cut a group's accuracy by 10% to 40% (Amayuelas et al., 2024). A recent collaborative version of debate raised weak-judge accuracy to 62.1%, from 49.2% with standard debate (Irving et al., 2018; Khan et al., 2024), but its judges accepted the models' consensus almost always, even when both were wrong, and it assumes honest models (Jiang et al., 2026). No existing debate holds up when agents share mistakes, lie, or collude.

4 Our approach: what is new, and why it will succeed

Our approach centers on debate, which narrows a disagreement to one step a judge can verify, so not every claim needs verifying (Figure 1). Each agent first answers on its own, with its evidence. Research goal 1 (RG1) weighs the answers by each agent's track record on planted cases (questions whose answers we already know) and picks debaters and judges impartially. If the weighted answers still disagree, the agents debate, and the judge's choice is the group's decision (RG2). Making this debate trustworthy is the core of the project: debaters must back claims with evidence the judge can re-run, and planted debates plus a few random audits by a person or a measurement catch debaters or judges who collude. RG3 trains agents to contribute evidence instead of winning debates.

RG1. Weigh agents, and pick debaters and judges impartially

Agents differ in skill, often by topic, and agents built on similar models make the same mistakes.

Task 1.1: Weights that count shared mistakes once. We will estimate each agent's track record (its accuracy, and which agents err with it) from planted cases and from how answers line up across real questions. The core method extends the rater model of Dawid and Skene (1979) with shared error factors: agents that keep getting the same cases wrong share a factor, and their agreement adds little evidence. In the vessel example, the two agents with the same outdated schedule count about once, so yes loses its weighted majority and the question goes to debate. Our PRD method weighted AI judges by peer ratings (Li et al., 2024a); here, planted cases anchor the weights.

Task 1.2: Impartial choice of debaters and judges. RG2 needs one representative per answer and a judge. Agents rate one another's arguments, as on our PeerEval leaderboard (peereval-ai.github.io), but an agent could then win a role by downgrading its rivals. In PRD, downgrading a rival also

lowered the rival's weight as a judge. We adapt the partition idea of impartial peer review (Alon et al., 2011; Kurokawa et al., 2015): agents built on the same base model form a group, and each group's members are ranked only by agents outside it. Each answer's representative is the top-ranked member of one group drawn at random from those that gave that answer, and the judge is an agent that did not answer and shares no base model with either side. An agent's ratings therefore never change its own selection. Where Lev et al. (2023) weight raters by agreement with the majority, we weight them by track record.

RG2. Settle disputes by agentic debate, and catch collusion

Text-only debate rewards persuasion, and debaters can collude to steer the conclusion.

Task 2.1: Agentic debate with re-runnable evidence. If the weighted answers still disagree, the agents debate in a shared sandbox where they can gather new evidence with tools such as search, code execution, and database queries. This builds on PRD's peer discussion and HiPRAG's search agents. Each answer sends one representative, chosen in Task 1.2, and the two walk through their arguments to the first step on which they differ (Irving et al., 2018). Every piece of evidence must carry the exact tool call that produced it. The judge, itself an agent with the same tools (Zhuge et al., 2025), re-runs the call, and the evidence counts only if the result matches. Debaters thus cannot invent or misquote evidence, and the judge rules on reproduced results, whereas judges in current tool-using debates weigh arguments (Jeong et al., 2026). The judge's choice is the group's decision. If no tool can settle the disputed step, the judge decides on argument alone and flags the decision as unverified. In the vessel example, the no side's query returns a port report, newer than the schedule, listing the vessel at its berth; the judge re-runs it, gets the same report, and picks no.

Task 2.2: Planted and audited debates against collusion. Collusion here means debaters, or the judge, steering a debate to the wrong side. A debater can argue its side badly on purpose, or colluders can mislead the judge with true but selectively chosen facts (Hu et al., 2026), which re-running evidence does not catch. We will mix in planted debates on planted cases, such as a paper with a deliberately inserted error, as in studies of human peer review (Schroter et al., 2008). Debaters and judges whose planted debates keep ending on the wrong side are flagged, and a monitor model that reads each transcript is vetted on planted examples of collusion, as in AI control (Greenblatt et al., 2024). Because capable models can often tell evaluations from real use (Needham et al., 2025), a colluder could behave well only in planted debates. We therefore audit a few real decisions chosen at random: a person (such as an analyst) or a direct measurement (such as a satellite image) finds the right answer. As with random tax audits, no agent knows which decisions will be audited, and agents that do worse on audited decisions than on planted ones are flagged.

RG3. Train agents that contribute evidence

Rewarding persuasion has produced agents that fabricate quotations (Kenton et al., 2026) and mislead people (Wen et al., 2025).

Task 3: Rewards for verified contributions. We will train agents with reinforcement learning on planted cases, rewarding each message that moves the group toward the right answer and penalizing each one that moves it away. This extends our process reward models, FG-PRM and HiPRAG (Li et al., 2025; Wu et al., 2026), from the steps of one agent to the messages of many. The main experiment compares training to contribute with training to win debates (Arnesen et al., 2024).

Evaluation: testbeds and a red team

All data are synthetic or public. We will use (A) math and code problems solved by agents of known skill, some sharing a flawed hint; (B) multi-document questions from our MEQA pipeline (Li et al., 2024b), including synthetic maritime scenarios; and (C) scientific claims and peer review, with reproducibility studies as audits. In the testbeds, an audit reveals the stored answer, within a fixed budget per method. Following AI control (Greenblatt et al., 2024), a red team will train debaters and judges to collude, to spot planted debates, and to cheat only when they expect no audit, and we will measure how many wrong decisions get through at each audit budget. Baselines are majority vote, rater weighting, our PRD, text-only and tool-using debate (Jiang et al., 2026; Jeong et al., 2026), and judge-weighting methods (Hossain et al., 2026; Zhu et al., 2026; Lev et al., 2023).

What is new. (i) A debate in which the judge re-runs the tool call behind every piece of evidence before it counts, so no side can win with invented or misquoted evidence. (ii) Planted debates and a few random audits that expose debaters or judges who collude, even those that behave well only in planted debates. (iii) Weights learned from planted cases, with agents that share mistakes counted about once. (iv) Rewards for moving the group toward the right answer, not for winning debates.

Why it will succeed. Theory supports the design: in debate, a judge who can verify one disputed step can settle arguments far beyond its own ability (Irving et al., 2018), and re-running evidence lets our judge verify that step. Without trusted audits, dishonest agents that imitate honest ones cannot be told apart from them, but a number of audits that does not grow with the number of items suffices, even when most agents are dishonest (Steinhardt et al., 2016). Most tasks build on the PI's own tools. For example, PRD agrees better with human judgments than the strongest single AI judge (Li et al., 2024a), and HiPRAG cuts unnecessary searches to 2.3% (Wu et al., 2026). The PI built PeerEval, the first leaderboard on which AI models rank one another, co-organized the AI4Research workshop at AAAI 2025, and co-wrote a survey of large language models for scientific research, including peer review (Luo et al., 2025). The PI also co-led one of five red teams in the Amazon Nova AI Challenge, holds a 2024 NSF CAREER award and a 2023 Amazon Research Award, and as a postdoctoral researcher led system integration for RESIN-11 in DARPA KAIROS.

5 Limitations of our approach

If no tool can settle a disputed step, the decision rests on argument alone, and we can only flag it. If every agent shares the same misconception, there is nothing to debate, and audits catch the error only by chance. A group that cheats only once, on a single high-stakes decision, may succeed before it is caught; we can bound such losses but not rule them out. Agents may also coordinate through hidden signals that monitoring misses (Mathew et al., 2025), so robustness is shown only for the attacks our red team tries. Operational Navy use would require Navy data and expert validation.

6 Impact if successful

The project will show when a debate among AI agents can be trusted and how few audits it takes to keep the agents honest. A Navy analyst using several AI tools would get a decision with its reasons: whose judgments carried weight and why, which evidence was reproduced, and whether the decision was flagged as unverified. These methods would let conferences and funders try impartial, audited peer review by AI. We will release code, testbeds, and protocols.

References

N. Alon, F. Fischer, A. D. Procaccia, and M. Tennenholtz (2011). Sum of Us: Strategyproof Selection from the Selectors. TARK, pp. 101–110.

A. Amayuelas, X. Yang, A. Antoniades, W. Hua, L. Pan, and W. Y. Wang (2024). MultiAgent Collaboration Attack: Investigating Adversarial Attacks in Large Language Model Collaborations via Debate. Findings of EMNLP, pp. 6929–6948.

S. Arnesen, D. Rein, and J. Michael (2024). Training Language Models to Win Debates with Self-Play Improves Judge Accuracy. arXiv:2409.16636.

H. K. Choi, X. Zhu, and S. Li (2025). Debate or Vote: Which Yields Better Decisions in Multi-Agent Large Language Models? NeurIPS.

A. P. Dawid and A. M. Skene (1979). Maximum Likelihood Estimation of Observer Error-Rates Using the EM Algorithm. Journal of the Royal Statistical Society, Series C (Applied Statistics) 28(1):20–28.

X. L. Dong, L. Berti-Equille, and D. Srivastava (2009). Integrating Conflicting Data: The Role of Source Dependence. PVLDB 2(1):550–561.

Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch (2024). Improving Factuality and Reasoning in Language Models through Multiagent Debate. ICML, PMLR 235:11733–11763.

B. Flanigan, P. Gölz, A. Gupta, B. Hennig, and A. D. Procaccia (2021). Fair Algorithms for Selecting Citizens' Assemblies. Nature 596:548–552.

S. Goel, J. Strüber, I. A. Auzina, K. K. Chandra, P. Kumaraguru, D. Kiela, A. Prabhu, M. Bethge, and J. Geiping (2025). Great Models Think Alike and this Undermines AI Oversight. ICML, PMLR 267:19621–19678.

R. Greenblatt, B. Shlegeris, K. Sachan, and F. Roger (2024). AI Control: Improving Safety Despite Intentional Subversion. ICML, PMLR 235:16295–16336.

E. Hossain, N. Yousefi, and S.-N. Lim (2026). Agreement Overstates Evidence: Error Dependence in LLM Judge Consensus. arXiv:2609.22512.

J. Hu, X. Huang, Y. Sun, Y. Dong, and X. Huang (2026). Lying with Truths: Open-Channel Multi-Agent Collusion for Belief Manipulation via Generative Montage. ACL.

G. Irving, P. Christiano, and D. Amodei (2018). AI Safety via Debate. arXiv:1805.00899.

A. Jaffe, E. Fetaya, B. Nadler, T. Jiang, and Y. Kluger (2016). Unsupervised Ensemble Learning with Dependent Classifiers. AISTATS, PMLR 51:351–360.

S. Jeong, Y. Choi, J. Kim, and B. Jang (2026). Tool-MAD: A Multi-Agent Debate Framework for Fact Verification with Diverse Tool Augmentation and Adaptive Retrieval. arXiv:2601.04742.

Y. Jiang, C. Chen, T. Wu, L. Sun, H. Liu, S. Feng, and C. Tan (2026). Collaborative Disagreement Resolution for Scalable Oversight. ICML.

Z. Kenton, L. Janzer, R. Greig, T. H. Teh, K. Tyshchuk, J. Brown-Cohen, H. Edwards, S. Rajamanoharan, N. Y. Siegel, N. Jaques, and R. Shah (2026). Debate Training Reduces Reward Hacking in RLAIF. arXiv:2608.17776.

A. Khan et al. (2024). Debating with More Persuasive LLMs Leads to More Truthful Answers. ICML, PMLR 235:23662–23733.

E. M. Kim, A. Garg, K. Peng, and N. Garg (2025). Correlated Errors in Large Language Models. ICML, PMLR 267:30038–30066.

G. Kohli (2026). Nine Judges, Two Effective Votes: Correlated Errors Undermine LLM Evaluation Panels. arXiv:2605.29800.

D. Kurokawa, O. Lev, J. Morgenstern, and A. D. Procaccia (2015). Impartial Peer Review. IJCAI, pp. 582–588.

O. Lev, N. Mattei, P. Turrini, and S. Zhydkov (2023). PeerNomination: A Novel Peer Selection Algorithm to Handle Strategic and Noisy Assessments. Artificial Intelligence 316:103843.

R. Li, T. Patel, and X. Du (2024a). PRD: Peer Rank and Discussion Improve Large Language Model based Evaluations. TMLR.

R. Li, Z. Wang, S. Q. Tran, L. Xia, and X. Du (2024b). MEQA: A Benchmark for Multi-hop Event-centric Question Answering with Explanations. NeurIPS 37, Datasets and Benchmarks, pp. 126835–126862.

R. Li, Z. Luo, and X. Du (2025). FG-PRM: Fine-grained Hallucination Detection and Mitigation in Language Model Mathematical Reasoning. Findings of EMNLP, pp. 4247–4278.

Z. Luo, Z. Yang, Z. Xu, W. Yang, and X. Du (2025). LLM4SR: A Survey on Large Language Models for Scientific Research. arXiv:2501.04306.

Y. Mathew, O. Matthews, R. McCarthy, J. Velja, C. Schroeder de Witt, D. Cope, and N. Schoots (2025). Hidden in Plain Text: Emergence and Mitigation of Steganographic Collusion in LLMs. IJCNLP-AACL, pp. 585–624.

J. Needham, G. Edkins, G. Pimpale, H. Bartsch, and M. Hobbhahn (2025). Large Language Models Often Know When They Are Being Evaluated. arXiv:2505.23836.

S. Schroter, N. Black, S. Evans, F. Godlee, L. Osorio, and R. Smith (2008). What Errors Do Peer Reviewers Detect, and Does Training Improve Their Ability to Detect Them? Journal of the Royal Society of Medicine 101(10):507–514.

X. Shi, Y. Zhang, and D. Yang (2026). Emergent Collusion in Long-Horizon LLM Agent Interaction. arXiv:2609.24967.

Y. Shu (2026). Unanimity Without Persuasion: A Single Round of Debate Erases the Disagreement That Verification Needs. arXiv:2609.26145.

J. Steinhardt, G. Valiant, and M. Charikar (2016). Avoiding Imposters and Delinquents: Adversarial Crowdsourcing and Peer Prediction. NeurIPS 29.

P. Verga et al. (2024). Replacing Judges with Juries: Evaluating LLM Generations with a Panel of Diverse Models. arXiv:2404.18796.

J. Wen, R. Zhong, A. Khan, E. Perez, J. Steinhardt, M. Huang, S. R. Bowman, H. He, and S. Feng (2025). Language Models Learn to Mislead Humans via RLHF. ICLR.

J. Whitehill, P. Ruvolo, T. Wu, J. Bergsma, and J. Movellan (2009). Whose Vote Should Count More: Optimal Integration of Labels from Labelers of Unknown Expertise. NeurIPS 22, pp. 2035–2043.

P. Wu, M. Zhang, K. Wan, W. Zhao, K. He, X. Du, and Z. Chen (2026). HiPRAG: Hierarchical Process Rewards for Efficient Agentic Retrieval Augmented Generation. ICLR.

L. Zheng et al. (2023). Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. NeurIPS 36, Datasets and Benchmarks, pp. 46595–46623.

B. Zhu, Y. Xie, and Y. Rao (2026). Stopping and Routing LLM Judge Panels. arXiv:2608.19802.

M. Zhuge, C. Zhao, D. Ashley, W. Wang, D. Khizbullin, Y. Xiong, Z. Liu, E. Chang, R. Krishnamoorthi, Y. Tian, Y. Shi, V. Chandra, and J. Schmidhuber (2025). Agent-as-a-Judge: Evaluate Agents with Agents. ICML.